

Lobito Corridor Analysis: A Configurable Decision Framework

Key Message

This is a **modular, scenario-based tool** – not a fixed recommendation. Results can be tailored to align with Angola's strategic priorities.

Equity-Heavy Scenario

- Higher weight on poverty & food insecurity
- Focus on rural poor, even if access gains are smaller

Balanced (Current)

- Balances need (poverty) and opportunity (access, population)
- Top 10% threshold with rural focus

Growth-Heavy Scenario

- Higher weight on travel time & cropland
- Focus on productive hinterlands & economic flows

This presentation shows results from the Balanced scenario. Different scenarios will produce different priority areas and recommendations.

Current Scenario: Balanced Configuration

All numbers in this presentation are based on the following parameter settings. These can be adjusted to test different policy priorities.

Priority Selection

- **Top 10%** of cells by composite score
- **Rural focus:** Require rural classification
- **Min cropland:** 5% threshold

Components Used

- ✓ Travel time to markets
- ✓ Population density
- ✓ Drought exposure
- ○ Nighttime lights (excluded for rural focus)

Equity Overlay Weights

- Poverty (W_POV): **15%**
- Food insecurity (W_FOOD): **10%**
- Travel time (W_MTT): **10%**
- Wealth index (W_RWI): **15%**

What Can Change?

Thresholds, weights, component mix, urban/rural split, and clustering parameters – all adjustable based on strategic priorities.

5 Strategic Questions This Framework Answers

Where do the strongest priority clusters emerge?

Where do multiple constraints stack with enough population to justify investment?

Are we focusing where needs and opportunities coincide?

Which municipalities balance high poverty with high potential for access improvements?

How many people benefit within 30/60/120 minutes?

Which sites reach the most beneficiaries quickly vs unlock remote hinterlands?

Are we stacking with other investments or creating islands?

Where can we coordinate with Government, World Bank and other partners?

How does movement along the corridor reinforce priorities?

Which segments carry the most interaction between priority areas?

Q1: Where do priority clusters emerge?

Under current configuration: Top 10% threshold, rural focus

- **Benguela:** 4 clusters, 432 km², 3,134 people (0.1% of province), 217 km² cropland
- **Huambo:** 2 clusters, 53 km², 31,776 people (1.1% of province), 35 km² cropland
- **Bie:** 1 cluster, 27 km², 18,307 people (0.9% of province), 11 km² cropland
- **Moxico:** 0 clusters
- **Moxico Leste:** 0 clusters

Key Finding

Huambo shows highest concentration: 1.1% of population in just 53 km² of priority clusters – indicates strong potential for targeted investment.

Moxico and Moxico Leste Challenge

Zero clusters means no areas cross current threshold. Options: (1) Lower threshold to 15%, (2) Adjust weights for Moxico and Moxico Leste context, or (3) Accept lower priority.

Q2: Are we focusing where needs and opportunities coincide?

Under current equity weights: W_POV=15%, W_FOOD=10%, W_MTT=10%, W_RWI=15%

Top 5 Municipalities - Huambo

1. Ekunha, **Score: 0.71**
2. Huambo, **Score: 0.70**
3. Caála, **Score: 0.67**
4. Bailundo, **Score: 0.62**
5. Katchiungo, **Score: 0.61**

Equity Quadrant Analysis

- **High Score & High Poverty, 3 municipalities (27.3%), 18% of rural poor:** Good alignment – invest here
- **High Score & Low Poverty, 3 municipalities (27.3%), 18% of rural poor:** Growth opportunities
- **Low Score & High Poverty, 3 municipalities (27.3%), 14% of rural poor:** At-risk of under-prioritization
- **Low Score & Low Poverty, 2 municipalities (18.2%), 50% of rural poor:** Lower urgency

Key Finding

Top municipalities combine high rural poverty (65-70%), large priority mask share (12-18%), and good catchment (40-46% within 60 min).

Scenario Sensitivity

Rankings shift with: (1) Higher poverty weight → rural-poor rise, (2) Higher access weight → better-connected rise, (3) Include urban → different set.

Q3: How many people benefit within 30/60/120 minutes?

Top Sites by 30/60/120-min Population (Huambo)

Site ID	Municipality	30-min	60-min	120-min
Site 1	Caála	24.3%	45.6%	67.9%
Site 2	Caála	34.7%	45.2%	68.8%
Site 20	Tchindjenje	14.7%	44.6%	68.2%
Site 5	Tchikala-tcholoanga	32.2%	43.5%	67.4%
Site 6	Tchikala-tcholoanga	19.9%	42.7%	64.4%

Quick Wins (30-60 min)

Caála sites (Sites 1 & 2) reach 45-46% of province population within 60 minutes – strongest immediate impact potential for market access and service delivery.

Remote Coverage (120 min)

All top 5 sites reach 64-69% of province population within 120 minutes – demonstrates solid network coverage for longer-distance agricultural supply chains and periodic services.

Strategic Trade-off

Investment choice: (1) **Concentrate in Caála** – Sites 1 & 2 reach 45% quickly with 30-min coverage of 24-35%, OR (2) **Distribute across 3 municipalities** - Adding Tchindjenje and Tchikala-tcholoanga sites improves geographic spread but with lower individual catchment percentages. Catchment tool can model combined coverage scenarios.

Q4: Are we stacking with other investments?

Project Density Around Clusters

Count of projects within 30 km of priority clusters (where available):

- **Huambo clusters:** Strong WB project density (Cluster 1: 23 projects, Cluster 2: 14 projects) - no Government or Other projects recorded
- **Benguela clusters:** No project location data available for synergy analysis, despite having 4 identified priority clusters

High-Opportunity Nodes

Clusters with many nearby projects → good bundling potential

Isolated Clusters

High need but few investments → risk of fragmentation

Bridge Opportunities

Areas that could connect separate project islands

Coordination Opportunity

Where WB and Government projects cluster within 30 km of priority areas, there's strong potential for: (1) Joint planning sessions, (2) Shared infrastructure (roads, electricity), (3) Coordinated service delivery. The tool can identify these overlap zones for each province.

Q5: How does movement along the corridor reinforce priorities?

Top Origin-Destination (OD) Pairs (Huambo)

- **Caála ↔ Huambo:** 97,939 flow, 56 km (✓ Both in priority mask)
- **Bailundo ↔ Huambo:** 55,540 flow, 97 km
- **Huambo ↔ Tchikala:** 49,511 flow, 36 km
- **Huambo ↔ Katchiungo:** 41,294 flow, 57 km
- **Ekunha ↔ Huambo:** 33,431 flow, 46 km

Key Finding

Caála-Huambo shows highest modelled flow (97,939) – **both ends are in priority mask**. This corridor segment deserves investment priority for both access and economic flow.

Gravity Model Assumptions

Flows based on population × distance decay. Currently using: OD_LAMBDA=0.015, RWI tilt enabled (RWI_WEIGHT=0.25). Can adjust to test different flow assumptions.

Strategic Insight

High-flow OD pairs that connect priority areas (like Caála-Huambo) should be prioritized for road/rail upgrades. These investments serve dual purposes: (1) improve access to priority clusters, AND (2) facilitate economic interaction between productive centers.

How Results Change Across Scenarios

Illustrative comparison: How would Huambo results differ under 3 scenarios?

Equity-Heavy

- 3-4 clusters
- Top: Ekunha (highest poverty)
- Pop in clusters: 1.2-1.4%
- Avg poverty: 72-75%
- Urban: No

Balanced (Current)

- 2 clusters
- Top: Ekunha
- Pop in clusters: 1.1%
- Avg poverty: 68-70%
- Urban: No

Growth-Heavy

- 1-2 clusters
- Top: Huambo city (best access)
- Pop in clusters: 0.8-1.0%
- Avg poverty: 60-65%
- Urban: Possible

Implementation Note

Any scenario can be run overnight by adjusting config.py parameters. Technical team needs 4-6 hours to reprocess all provinces.

Decision Point

Which scenario best aligns with Angola's corridor strategy? We can run 2-3 scenarios in parallel to compare trade-offs before final investment decisions.

Illustrative projections based on parameter sensitivity. Actual scenario runs would provide precise numbers for all 5 provinces.

Recommendations & Next Steps

Immediate (1-2 weeks) ?

- Validate current Balanced scenario alignment
- Decide on Moxico and Moxico Leste: accept zero clusters OR adjust threshold
- Additional data: Project locations from WB, Govt, Others, for all provinces within corridor (Currently Huambo and Moxico are the only province has WB project locations.

Medium-term (1-2 months)

- Run 2-3 comparison scenarios
- Co-design with transport, agriculture, energy teams
- Enrich project metadata by sector

Longer-term (3-6 months)

- Integrate observed data (traffic, production, trade)
- Add remaining provinces to analysis
- Template for other Angola corridors

Discussion: Which scenario should we prioritize?